

The minimal even icosahedral Artin representation: computational results and a catalogue of obstructions

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Abstract

Artin’s holomorphy conjecture is open in exactly one case for $GL_2/\mathbb{Q}$: even representations with projective image A_5 . By a theorem of Booker, Lee and Strömbergsson, conditional on Artin’s conjecture the smallest conductor among these is 1951, realised by a representation of Doud and Moore. We report a computational study of that representation. We verify its defining field ($d_K = 1951^4$, A_5 , totally real, $e = 5$ at 1951), and we establish that the adjoint $\text{Ad } \rho$, a three-dimensional representation of A_5 requiring no lift to $2.A_5$, has an L -function satisfying its functional equation at conductor 1951^2 with root number $+1$ to 10^{-6} , against $\geq 10^{-2}$ for every competing conductor, with 11 zeros on the critical line for $0 < t \leq 6$ against a predicted 11.1007. Since A_5 has no subgroup of index three, $\text{Ad } \rho$ is not monomial and its holomorphy is open; this appears to be the first numerical evidence for it at the minimal conductor.

We then catalogue, with a validated instrument, what does *not* work. The family $a_p = \eta(p)\chi(p)^k \varepsilon_p |t_p|$ arising from Crespo’s solution of the embedding problem $A_5 \hookrightarrow 2.A_5$ is exhausted: all twists, both parities, all eight admissible quadratic classes, and all sign configurations reachable by coordinate descent from three independent starts. The residuals are structural, generalising to held-out evaluation points at ratio 0.9–1.1. The instrument itself is validated at 3.4×10^{-30} on Eisenstein series. Nine earlier verdicts in our own working record are shown to have been void, each for a specific and reproducible reason, and we give the corrected statements. Finally we report a pure-Python port of the numerical core of Lowry-Duda’s Hejhal implementation, gated at level one to $R = 9.533695261$ with $a_2 = -1.06833355$ and $a_3 = -0.45619735$, and we identify the single remaining modification — nebensystem of order five — separating it from the target.

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1 Introduction

1.1 The open cell

Let $\rho: G_{\mathbb{Q}} \rightarrow \mathrm{GL}_2(\mathbb{C})$ be irreducible with finite projective image, necessarily dihedral, A_4 , S_4 or A_5 . Write $\rho(c)$ for complex conjugation; ρ is *odd* if $\det \rho(c) = -1$ and *even* otherwise. The status of Artin holomorphy is:

	solvable image	A_5
odd	Langlands–Tunnell	theorem [1, 8]
even	Maass [10], Langlands–Tunnell	open

The odd icosahedral case has the same group, the same failure of the radical tower, and the same field $\mathbb{Q}(\sqrt{5})$, and it is a theorem. So non-solvability alone does not explain the open cell. The obstruction is the conjunction of two independent conditions: the image is not solvable, so cyclic base change does not reach it; and the archimedean type is even, so the corresponding automorphic object is a Maass form of Laplace eigenvalue $1/4$, which is not cohomological and admits no p -adic deformation over $\mathbb{Q}$. Drop either condition and the cell closes. Calegari [5] states the consequence plainly: Artin’s conjecture is not known for a single representation with image the binary icosahedral group.

1.2 Why conductor 1951

Booker, Lee and Strömbergsson [2], using a twist-minimal trace formula, classify even two-dimensional Artin representations of conductor at most 2862 conditional on Artin’s conjecture. The minimum is 1951, attained by a representation found by Doud and Moore [7]. Their type 3a has $e_p = 5$ and prime conductor, which forces $p \equiv 1 \pmod{5}$: conductor exponent one requires inertia with a fixed vector, so $\rho|_{I_p} = \mathrm{diag}(1, \chi_5)$ and $\det \rho|_{I_p}$ has order five, whence $5 \mid p - 1$. All six conductors in their table — 1951, 2141, 3701, 3821, 8501, 9461 — satisfy $p \equiv 1 \pmod{5}$. Types 3b and 3c instead have $d_K = p^2$; the smallest occur at $p = 10267$ and $p = 13613$.

Conductor governs every computational route. The base change of ρ to an imaginary quadratic field, which is where the representation becomes cohomological in the sense of Calegari–Geraghty, has level norm $\mathrm{cond}(\rho)^2$. For 1951 that is 3.8×10^6 , at the edge of Bianchi modular symbols; for a

conductor-15733² representation it is 6.1×10^{16} , which is hopeless. Minimal conductor is therefore not an aesthetic preference but the only regime in which the analytic methods operate at all.

1.3 Results

Theorem 1.1. *The polynomial $f(x) = x^5 - x^4 - 780x^3 - 1795x^2 + 3106x + 344$ defines a totally real quintic field K with $\text{Gal}(\tilde{K}/\mathbb{Q}) \cong A_5$, $d_K = 1951^4$, and 1951 totally ramified ($e = 5$, $f = 1$). Its index is $136178 = 2 \cdot 7 \cdot 71 \cdot 137$. The Serre obstruction to $A_5 \hookrightarrow 2.A_5$ vanishes: the trace form is $\mathbb{Q}$ -equivalent to $\langle 1, 1, 1, 1, 1 \rangle$, with an explicit isometry P satisfying $P^\top TP = I$ and $\det P = 1/1951^2$.*

Theorem 1.2. *Let $\text{Ad } \rho = \text{Sym}^2 \rho \otimes (\det \rho)^{-1}$, a three-dimensional representation of the projective image A_5 , computable from K with no lift. Then $\text{cond}(\text{Ad } \rho) = 1951^2$, the archimedean factor is $\Gamma_{\mathbb{R}}(s)^3$, and*

$$\Lambda(s) = (1951^2)^{s/2} \Gamma_{\mathbb{R}}(s)^3 L(s, \text{Ad } \rho) = +\Lambda(1-s)$$

to relative error 10^{-6} , against $\geq 2 \times 10^{-2}$ for the competing conductors 1951, 1951^3 , 1951^4 and for root number -1 . Furthermore $\Lambda(1/2 + it)$ has exactly 11 sign changes for $0 < t \leq 6$ against $\theta(6)/\pi = 11.1007$, with $S(T)$ oscillating in $[-0.41, +0.50]$ and no drift. Since A_5 has no subgroup of index three, $\text{Ad } \rho$ is not monomial and its holomorphy is not known.

Theorem 1.3. *Write $a_p = \eta(p)\chi(p)^k \varepsilon_p |t_p|$, where $|t_p|$ is determined by the A_5 -class of Frob_p , χ is the order-five nebentypus, η is quadratic, and $\varepsilon_p = \pm 1$ is the lift datum supplied by Crespo's $\gamma = \det(MP + I)$. No member of this family satisfies*

$$f|_0 W_N = \lambda_N f^*, \quad \lambda_N = \tau(\chi^{2k})/\sqrt{N},$$

at any of: both independent twists k (note $k \leftrightarrow 5-k$ are conjugate), both parities, all eight quadratic classes η admissible for conductor 1951, both free index-divisor signs, and all sign configurations reachable by coordinate descent over the 74 signs live at $M = 600$ from Crespo's configuration and from random starts. The best residual is 0.038; even-parity descent reaches exact floors 0.794104 and 0.522511 identically from all starts. All configurations generalise to held-out evaluation points at ratio 0.9–1.1, so the residuals are structural rather than fitted.

1.4 What this paper is

Two of the three theorems are negative or infrastructural. We think this is the useful contribution. A large part of the effort in this area consists of tests that cannot distinguish a correct candidate from an incorrect one, and our own record contains nine such. Section 5 lists them with the reason each failed, because a catalogue of instruments that do not work is worth as much to the next attempt as a catalogue of objects that do.

2 The field and the lift

2.1 Verification

All of Theorem 1.1 is checked in PARI/GP [11]: $\text{disc}(f) = 2^2 \cdot 7^2 \cdot 71^2 \cdot 137^2 \cdot 1951^4$, $\text{nfdisc} = 1951^4$, $\text{polgalois} = [60, 1, 1, A_5]$, $\text{polsturm} = 5$, and idealprimedec gives $e = [5]$, $f = [1]$ at 1951.

The index divisors 2, 7, 71, 137 matter. At these primes $f \bmod p$ is not squarefree, $\mathbb{Z}[\theta]$ is not maximal, and any computation of the Frobenius class from the factorisation of $f \bmod p$ returns a

wrong answer *silently*: the Frobenius-orbit computation takes place in a ring that is not étale. We record the correct data from `idealprimedec`:

$$2, 137 : (3, 1, 1), \quad 7, 71 : (2, 2, 1).$$

In our first pass all four were assigned the class (5) with $a_p = \varphi$; correcting them moved a functional-equation residual from 5.7×10^{-2} to 5×10^{-6} , which is the difference between a failed test and a passed one.

2.2 The Serre obstruction

Let $T = [\text{Tr}(u_i u_j)]$ be the Gram matrix of the trace form on the `nfin` integral basis. Serre's formula [12] gives $w_2(Q_K) = sp(K) + (2) \cdot (d_K)$; as d_K is a square the second term vanishes. Solving $q(w) = 1$ repeatedly by `qfsolve` on $q \perp \langle -1 \rangle$ and splitting off orthogonal complements produces

$$P^\top T P = I_5 \text{ exactly,} \quad \det P = \frac{1}{1951^2},$$

so $Q_K \cong \langle 1, 1, 1, 1, 1 \rangle$ over $\mathbb{Q}$ and $sp(K) = 0$. The lift exists.

Remark 2.1. The relevant decomposition is $\rho = \rho_0 \otimes \chi^3$, where χ is the order-five nebentypus and $3 \cdot 2 \equiv 1 \pmod{5}$, so $\det \rho_0 = \chi^6 = 1$ and ρ_0 lands in $\text{SL}_2(\mathbb{F}_5)$. Crespo's construction is written for exactly that situation. Without this observation the GL_2 problem looks inaccessible to Crespo; with it, the missing datum is the sign ε_p of an $\text{SL}_2(\mathbb{F}_5)$ lift on the same quintic.

2.3 Crespo's construction and its orientation

Crespo [6] (see also Bryk [4]) gives $\gamma = \det(MP + I)$ with M the embedding matrix. We record a degeneracy not stated in the sources we consulted.

Lemma 2.2. $M^\top M = T$ and $P^\top T P = I$ give $(MP)^\top (MP) = I$, so MP is orthogonal. In odd dimension an orthogonal matrix of determinant -1 has -1 as an eigenvalue, so $\det(MP + I) = 0$ identically.

Since $\det M = \pm \sqrt{\text{disc } f}$ according to the parity of the root ordering, exactly half of all orderings annihilate γ ; we observed this as a 50% failure rate before identifying the cause. Transposing two roots restores $\det(MP) = +1$, and the remaining orderings differ by even permutations, that is by elements of the Galois group. Because $\gamma^{\sigma^{-1}}$ is a square, all conjugates lie in one square class, so $\gamma \bmod \mathfrak{P}$ may be built directly in the residue field and the global element is never needed.

3 The adjoint L -function

$\text{Ad } \rho$ is a representation of the *projective* image, so it is computable from the quintic alone: no lift, hence no dependence on ε_p , hence none of the obstructions of §4. Tame inertia of order five acts on the three-dimensional space with eigenvalues $1, \zeta, \zeta^{-1}$, leaving one fixed vector, so the conductor exponent is $3 - 1 = 2$ and $\text{cond}(\text{Ad } \rho) = 1951^2$. Since $\rho(c) = \pm I$ acts trivially on the adjoint, the archimedean factor is $\Gamma_{\mathbb{R}}(s)^3$.

3.1 The test

With $\int_0^\infty k(x)x^{s-1}dx = \Gamma(s/2)^3$ one obtains

$$\Lambda(s) = \int_0^\infty \mathcal{K}(v)v^s \frac{dv}{v}, \quad \mathcal{K}(v) = \sum_{n \geq 1} a_n k\left(\frac{\pi^{3/2}nv}{\sqrt{N}}\right),$$

and the functional equation becomes the pointwise identity $\mathcal{K}(v) = w\mathcal{K}(1/v)/v$. The kernel is $k(x) = 2G_{0,3}^{3,0}(x^2 \mid 0, 0, 0)$, tabulated by `mpmath` on 4000 log-spaced points with the Mellin identity verified to 3×10^{-10} . (The analogous degree-four kernel is the degenerate logarithmic case and `mpmath` fails on it; we obtained that one by contour integration.)

3.2 Result

conductor	w	relative error
1951 ²	+1	5×10^{-6}
1951	+1	2.2×10^{-1}
1951 ³	+1	3.8×10^{-2}
1951 ⁴	+1	7.1×10^{-1}
1951 ²	-1	2.0

The zero count, from $\Lambda(1/2 + it) = 2 \int_0^\infty \mathcal{K}(e^u)e^{u/2} \cos(tu) du$ with half weight at $u = 0$:

T	zeros	$\theta(T)/\pi$	$S(T)$
2	2	2.4110	-0.4110
4	7	6.5039	+0.4961
6	11	11.1007	-0.1007

$\Lambda(1/2) = 1657.77$. Every zero located is a sign change of a real function on the critical line, and the count matches the argument principle, so no zero lies off the line and no pole is present for $0 < t \leq 6$.

Remark 3.1 (Why the range stops at 6). $|\Lambda(1/2 + it)| \sim e^{-3\pi t/4}$ for degree three, so at $t = 8$ the signal is 1.1×10^{-5} absolute against $\Lambda(1/2) = 1657.77$. A relative floor of 10^{-6} on $\mathcal{K}$ is 8×10^{-4} absolute and exceeds the signal at $t \approx 7$. We verified that this floor is *not* the kernel: upgrading it from 10^{-6} to 3×10^{-10} changed neither the functional-equation residual nor the zero count. Nor is it spline error (10^{-8}), roundoff (10^{-12}) or series truncation (e^{-81}). The floor grows with v , from 1.0 to 6.1×10^{-6} over $v \in [1.06, 1.3]$, so it is not a constant misnormalisation either. We have not identified it.

4 The exhausted family

4.1 What is being tested

For a weight-zero form the slash operator carries no automorphy factor, and W_N normalises $\Gamma_0(N)$ while conjugating the character:

$$W_N : L^2(\Gamma_0(N) \backslash \mathbb{H}, \chi) \longrightarrow L^2(\Gamma_0(N) \backslash \mathbb{H}, \bar{\chi}).$$

For χ of order five there is therefore *no* scalar λ with $f|W_N = \lambda f$. The correct relation is $f|_0 W_N = \lambda_N f^*$ with f^* carrying $\overline{a_n}$. The constant is the root number: at infinity $\rho(c) = +I$ gives $W_\infty = 1$, and at N the tame type $\rho|_I = \text{diag}(1, \chi)$ gives $\tau(\chi)/\sqrt{N}$, with no i^{-k} because $k = 0$. Numerically $\lambda_N = -0.88903855692463 - 0.45783233208389i$, computed three independent ways: the holomorphic convention, the local root number, and the functional equation $\Lambda(s, \chi) = (\tau/\sqrt{N})\Lambda(1-s, \overline{\chi})$ of the Dirichlet L -function, which we verified to 10^{-37} .

4.2 The scan

Theorem 1.3 reports the outcome. Two implementation traps are worth recording because each corrupts every even coefficient and neither is visible without a control. The Kronecker symbol at $n = 2$ is not zero for odd fundamental discriminant d : $(d/2) = +1$ if $d \equiv \pm 1 \pmod{8}$ and -1 if $d \equiv \pm 3 \pmod{8}$. And at $p \mid d$ the twist is ramified, so $a_{p^k} = 0$; feeding $a_p = 0$ into the unramified Hecke recursion yields $a_{p^2} = -1$ instead. A $d = 1$ control isolates both.

4.3 The instrument

Before any of this is meaningful the tester must be validated. We use the real analytic Eisenstein series, whose K_{ir} expansion is generated from ξ , ζ and $\sigma_a(n)$ in-script and which satisfies $E(-1/z, s) = E(z, s)$ exactly. Over nine rows — three spectral parameters including $r = 9.5337$ and $r = 0.5$, three points each — the worst relative deviation is 3.4×10^{-30} . This tests complex Bessel order, the $\sqrt{y}$ factor, the cosine sum, the pairing across heights differing by a factor of five, and the unnormalised residual convention, using no imported coefficients.

5 Void verdicts

The following appeared in our working record as settled and were not.

1. **Fricke at $y^* = 1/\sqrt{N}$.** That point is the fixed point of W_N . The relation degenerates to $f(z^*)(1 - \lambda_N) = 0$ and the residual is the constant $|1 - \lambda_N|/2 = 0.971864$ for any input. Four experiments reported residuals near 0.97.
2. **Normalised comparison.** Rescaling f at each height discards the magnitudes, which are the entire content of the relation, and returns the same constant. This is why conjugating the stream “did not move a digit”.
3. **$f|W = \lambda f$ in the same space.** Wrong equation for non-real χ , as above.
4. **One λ_N for every twist.** $\rho_0 \otimes \chi^k$ has nebentypus χ^{2k} , so λ_N must be $\tau(\chi^{2k})/\sqrt{N}$. A scan over k against a single λ_N tests wrong relations.
5. **Odd parity on the imaginary axis.** An odd form vanishes there identically; every such residual compared 0 with 0.
6. **“Crespo fails because the signs are not on $\mathcal{O}_K$ ”.** The obstruction vanishes and the signs are computable; the correct statement is that the resulting stream fails a tester that can read it.
7. **The S_2 congruence probe.** Structurally empty: $\overline{\rho}$ is even, while every weight-two mod- λ representation is odd for every nebentypus and level, and parity is a twist invariant.

8. **Row scaling in the linear system.** Dividing row m by d_m amplifies high- n rows by $e^{2\pi m(y^* - Y)}$, giving $\kappa = 1.8 \times 10^{38}$; the smallest singular value is then roundoff at every r . The column scaling $c_n = a_n d_n$ gives $\kappa = 1.016$, and is valid only when $Y < \min_j y_j^*$.
9. $Q > M$. The trigonometric projection has rank at most Q per cusp, so $M > Q$ manufactures $2(M - Q)$ spurious null directions at every r .

We also record a near-miss worth naming. Searching the *even* sector near $R = 9.53$ produces an argmin at 9.5385, within 5×10^{-3} of the first eigenvalue — which is *odd*. The coefficients settle it ($a_2 = 0.185$ against -1.068), but the eigenvalue alone does not. A flat minimum of a featureless curve is still a minimum.

6 A gated solver

The construction of the Maass form requires automorphy under many elements of $\Gamma_0(1951)$, not one Atkin–Lehner involution. We report a pure-Python port of the numerical core of Lowry-Duda’s `heuristic_hejhal` [9], which clears the level-one gate:

$$R = 9.533695261 \text{ (err } 3.5 \times 10^{-10}), \quad a_2 = -1.06833355, \quad a_3 = -0.45619735,$$

all eight digits of the published values. Two features of the reference are essential and were absent from our own earlier assembly.

First, there is *one* coefficient stream, not one per cusp: the cusp enters only through $\text{signs}[\text{cusp}] = \pm 1$ and the local coordinates, which is the Atkin–Lehner structure. Solving for independent streams at each cusp yields a system degenerate at every r .

Second, the eigenvalue detector is a Hecke relation, $C(R) = a_2 a_3 - a_6$, not agreement between two sampling heights. The relation is not imposed by the solver, so its vanishing is independent evidence. Our own level-one solver produced $a_2 a_3 = a_6$ to 8×10^{-10} unbidden before we recognised it as the detector.

Separately we verified the $\Gamma_0(N)$ pullback on 4000 random points with no failures ($\det \gamma = 1$, $c \equiv 0 \pmod N$), and measured the height distribution: at $N = 1951$ the median pullback height is 2.9×10^{-6} and 85% lie below 10^{-2} , so a K -Bessel sum at ∞ would need n up to 1.6×10^6 at the median. The cusp-zero expansion is not an optimisation but a requirement. We validated it against the level-one form viewed as an oldform at level five, where $b_m = \sqrt{N} a_{m/N}$ for $N \mid m$ is exact, to 6×10^{-10} .

Remark 6.1. A sign error cost us a pass here: $f(-1/w) = f(w)$ for *both* parities, because $S \in \text{SL}_2(\mathbb{Z})$. The odd/even split is the involution $z \mapsto -\bar{z}$. Conflating them produces magnitudes agreeing to nine digits with opposite signs.

The remaining modification is nebentypus of order five: the reference handles trivial character only. This makes the coefficients complex, replaces $\sin / \cos$ by $e(nx)$, doubles the unknowns to $\pm n$, and replaces $\text{signs}[\text{cusp}]$ by $\chi(d_\gamma)$, which the verified pullback supplies. This is an extension of published code, not a parameter change.

7 Limitations

- Nothing here constructs a Maass form, and nothing bears on Artin’s conjecture in general. Theorem 1.2 is numerical evidence for one L -function to finite height, in the tradition of [3].

- The conductor determinations are numerical. The zero counts are conditional on an explicit Turing bound on $\int S(t) dt$, which we do not supply; the numerical margin is a factor of 5 to 20.
- Theorem 1.3 exhausts a parametrised family, not all streams. ε has 2^{430} configurations; Crespo's γ picks one and its quadratic torsor eight, and coordinate descent explores a neighbourhood. Distinguishing the remainder requires a solver that treats the coefficients as unknowns.
- The floor of 10^{-6} in §3 is unexplained.
- The scattering matrix of $\Gamma_0(1951)$ with χ_2 , whose trace at $s = 1/2$ enters the parabolic term of the Selberg trace formula and bears on the multiplicity of $\lambda = 1/4$, we could not compute: two candidate forms for $\varphi_{\infty 0}$ fail unitarity, and for complex χ the condition is $\Phi(s)\Phi(1-s) = I$ across the conjugate pair rather than $|\varphi| = 1$. This needs the explicit Kloosterman sum.

Data and code

Defining polynomials, integral bases, orthonormalisers, coefficient files, the certified degree-three kernel, the validated Eisenstein control, the pullback, and the ported solver are available from the authors. Computations used PARI/GP 2.15.4 [11] and Python with NumPy, SciPy, SymPy and mpmath.

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